

Comparative Analysis & Discussion Writing

Comparison with literature, interpretation of findings, and effective discussion writing

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1

Compare

Place your findings beside previous studies.

2

Interpret

Explain what the findings mean.

3

Discuss

Connect results with gap, scope, and contribution.

After this lecture, students will be able to:

01

Outcome 01

Explain the purpose of comparative analysis in research writing.

02

Outcome 02

Prepare a literature comparison table using relevant studies.

03

Outcome 03

Interpret research findings in relation to objectives and research questions.

04

Outcome 04

Write a structured discussion section with limitations and implications.

Why do we compare our findings with literature?

A

Positioning

Shows where your study stands compared with existing knowledge.

B

Validation

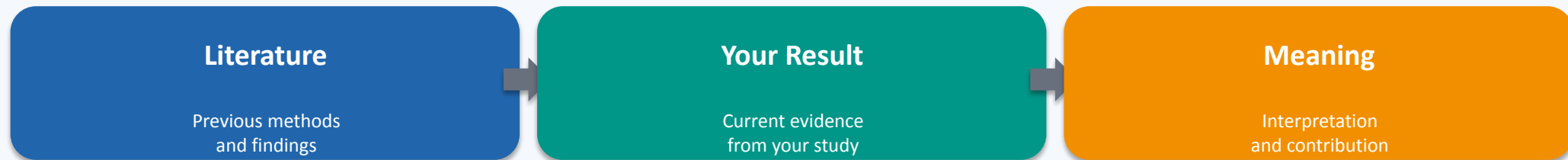
Checks whether your results agree with or differ from previous findings.

C

Contribution

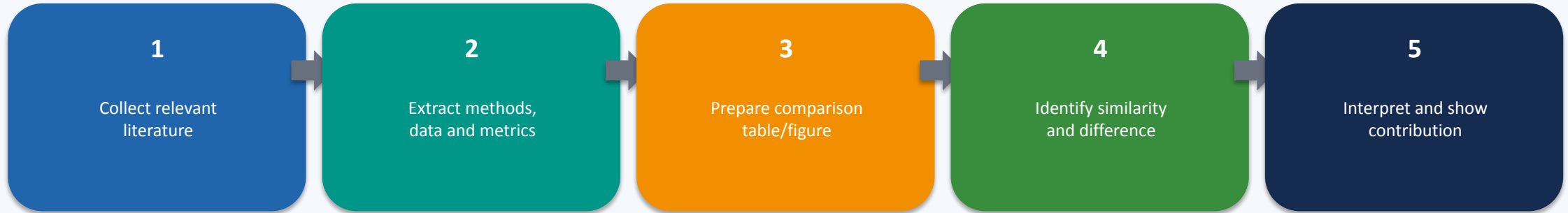
Clarifies what is new, improved, or context-specific in your work.

A good comparison does not only say “our result is better.” It explains why the result is similar, different, improved, limited, or meaningful.



Results Section	Discussion Section
Presents what was found	Explains what the findings mean
Uses figures, tables, metrics, observations	Uses interpretation, comparison, implication
Mostly objective and direct	Analytical and explanatory
Answers: What happened?	Answers: Why does it matter?
Avoids excessive explanation	Connects findings with literature and objectives

Key message: Results show evidence. Discussion explains value.



Quantitative comparison

Compare numerical results such as accuracy, F1-score, error rate, cost, time, or efficiency.

Qualitative comparison

Compare methods, assumptions, datasets, tools, scope, limitations, and practical relevance.

Contextual comparison

Explain why results may differ due to country, sample, environment, domain, or implementation.

Do not compare only with the highest-performing study. Compare with the most relevant studies.

Selection Point	What to Check
Relevance	Same topic, problem, method, dataset, variable, or application area
Recency	Recent papers, preferably from the last 3–5 years where appropriate
Quality	Indexed journals, reputable conferences, systematic reviews, benchmark studies
Similarity	Comparable population, dataset, context, sample size, or evaluation metric
Diversity	Include studies with similar, better, weaker, or different findings

Practical Tip

Prepare a matrix before writing: Author, year, method, data, metric, result, limitation, and your comparison point.

Study	Method	Dataset / Sample	Metric	Key Result	Comparison Point
Author A (2024)	CNN	1,200 images	Accuracy	91.4%	Lower than proposed model
Author B (2023)	Random Forest	Hospital records	F1-score	0.86	Similar trend, smaller dataset
Proposed Study	Optimized ML	2,500 records	F1-score	0.91	Improved performance and better balance

How to discuss the table

- First, state the overall trend.
- Then compare your result with the most relevant studies.
- Explain possible reasons for differences.
- End by linking the comparison to your contribution.

Avoid this mistake

Do not paste a table without explaining what it shows. The table supports the writing, but it does not replace the discussion.

Useful sentence pattern

Pattern 1: Agreement

The findings are consistent with [Author], who reported that...

Pattern 2: Difference

This result differs from [Author], possibly because...

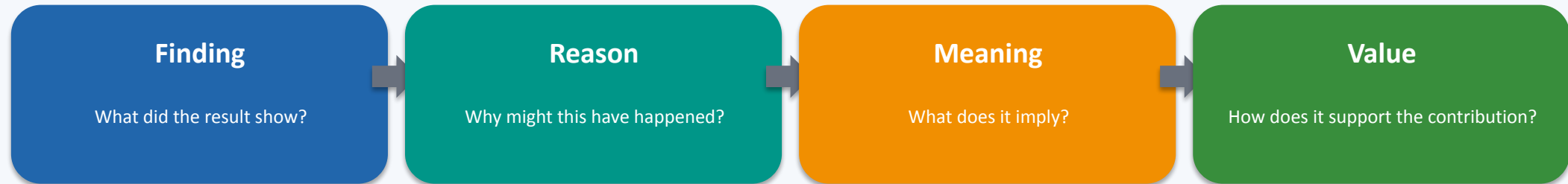
Pattern 3: Improvement

Compared with previous studies, the proposed approach achieved...

Mini Example

The proposed model achieved a higher F1-score than earlier studies using conventional machine learning. This improvement may be due to balanced data preprocessing, optimized hyperparameters, and the use of more representative features. However, direct comparison should be interpreted carefully because the datasets and validation settings were not identical.

Interpretation explains the meaning behind the result



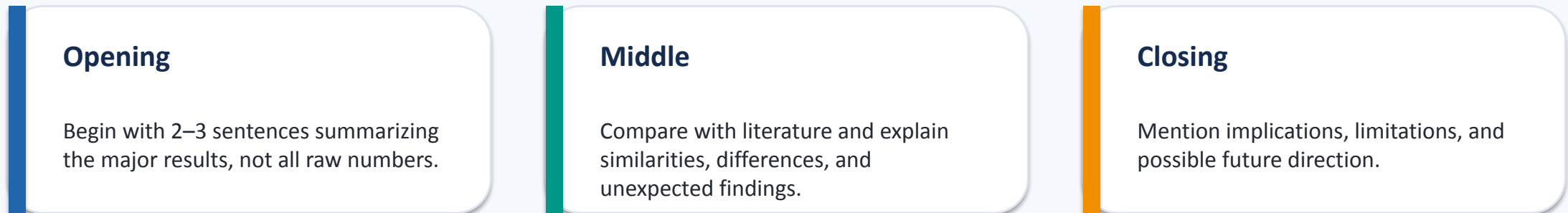
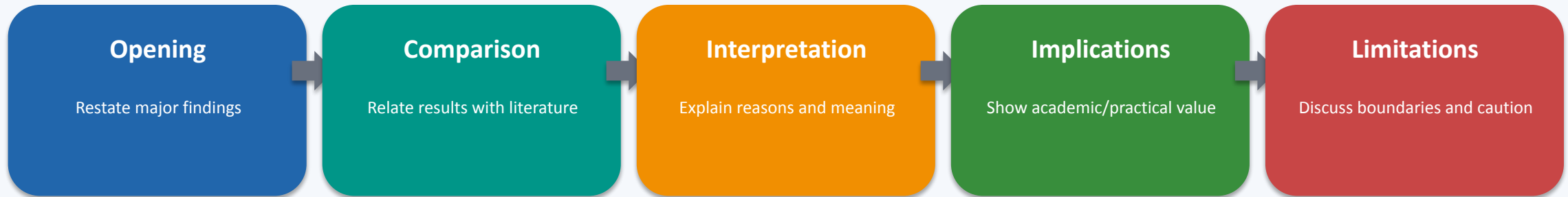
Weak interpretation

The result was good and better than previous studies.

Strong interpretation

The result indicates that feature selection reduced noise and improved model generalization, especially for minority-class prediction.

A strong interpretation connects evidence, reason, and research objective.



Part	Purpose	Example Starter
Topic sentence	State the main finding	The results indicate that...
Comparison	Link with literature	This finding is consistent with...
Explanation	Explain why	A possible reason is...
Implication	Show meaning	This suggests that...
Limitation / caution	Avoid overclaiming	However, this result should be interpreted carefully because...

Paragraph Rule

One paragraph should normally discuss one key finding, one comparison point, and one interpretation.



When findings are similar

Explain how your result supports previous evidence. Mention whether it confirms, extends, or strengthens earlier findings.



When findings are different

Explain possible causes such as dataset, method, sample size, measurement tool, time period, or context.

Useful explanation points

- Different data source or sample characteristics
- Different preprocessing or measurement approach
- Different model, tool, or experimental setup
- Different geographic, social, or institutional context
- Different evaluation metric or validation strategy

Strong discussion is balanced, not exaggerated

Weak Claim	Improved Claim
This model is the best model.	The model achieved the highest score among the compared methods in this experimental setting.
This study proves that the method works everywhere.	The findings suggest that the method is promising for the selected dataset and context.
The result is completely accurate.	The result shows acceptable performance, though further validation is needed.
No previous study has done this.	Limited studies have addressed this issue in the selected context.

Tip

Use cautious language: suggests, indicates, may be due to, appears to, in this context, based on the findings.



Limitations

Mention the boundaries of your study honestly: sample size, dataset quality, method, assumptions, time, scope, or generalizability.



Implications

Explain how your findings may help researchers, practitioners, policymakers, institutions, industry, or future studies.

Type	Example
Practical implication	The method may support faster decision-making in real settings.
Theoretical implication	The finding extends existing understanding of the relationship between variables.
Methodological implication	The proposed framework can guide future model development.
Limitation	The dataset came from a single source, so broader validation is needed.

Mistake	Better Practice
Repeating results only	Explain meaning and connect with literature
No comparison with literature	Compare with relevant, recent, and credible studies
Ignoring contradictory findings	Discuss differences and possible reasons
Overclaiming contribution	Use evidence-based and cautious statements
No limitation section	Mention limitations honestly and briefly
Unorganized paragraphs	Use one paragraph for one key discussion point

A discussion section should be analytical, evidence-based, and connected to the research objectives.

Example

The proposed approach achieved improved classification performance compared with the baseline models. This finding is consistent with previous studies that reported performance gains after applying feature selection and parameter optimization. The improvement may be explained by the reduction of irrelevant features, which helped the model focus on more informative patterns. However, the result should be interpreted with caution because the evaluation was conducted on a limited dataset. Future studies may validate the approach using larger and more diverse samples.

Finding

Comparison

Reason

Caution

☐ Have I restated the major findings briefly?

☐ Have I compared my findings with relevant literature?

☐ Have I explained similarities and differences?

☐ Have I interpreted the meaning of the findings?

☐ Have I linked the discussion to objectives or research questions?

☐ Have I included limitations and implications?

☐ Have I avoided unsupported claims?

Comparison gives context.
Interpretation gives meaning.
Discussion gives value.

1

Compare

What did others find?

2

Interpret

Why did this happen?

3

Contribute

What value does your study add?

A strong discussion section shows that the researcher understands both the result and the wider research field.